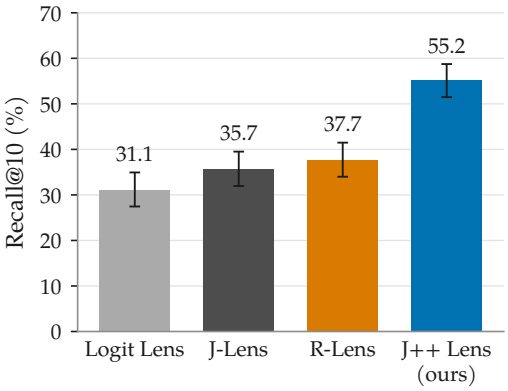


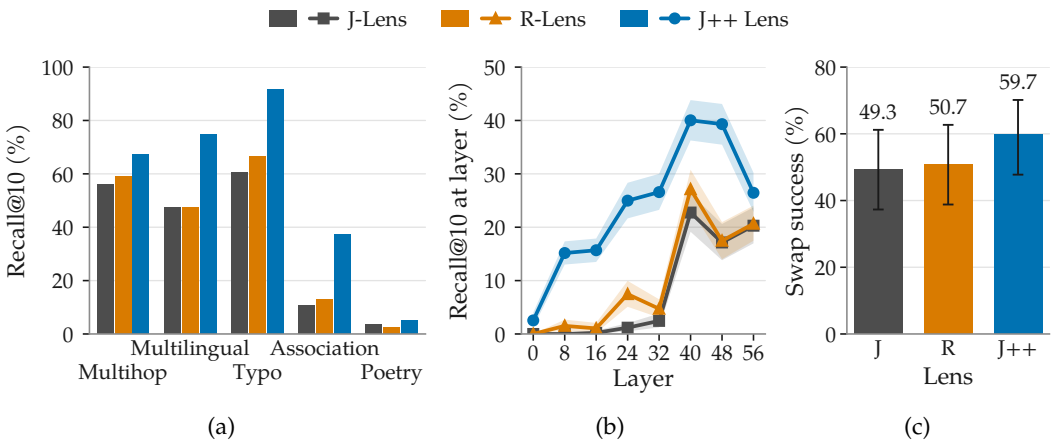
(a)

Fact: In humans, the organ that pumps blood through the body has this many chambers:

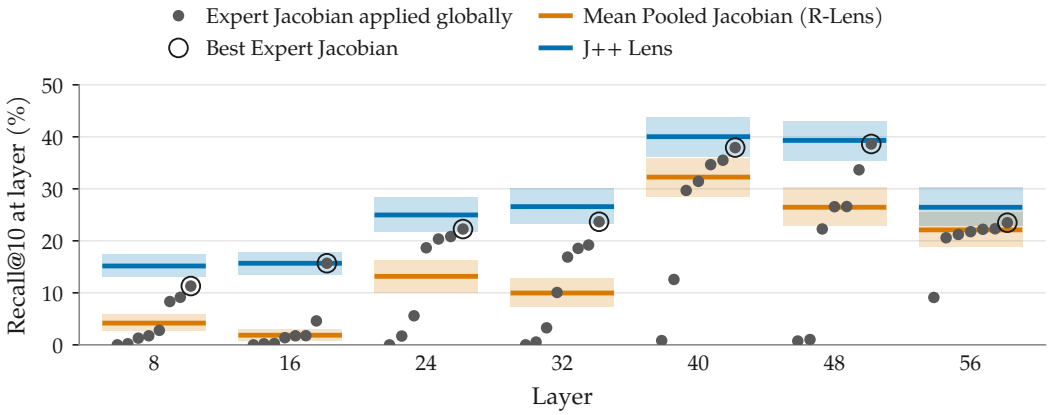
heart target related to the target

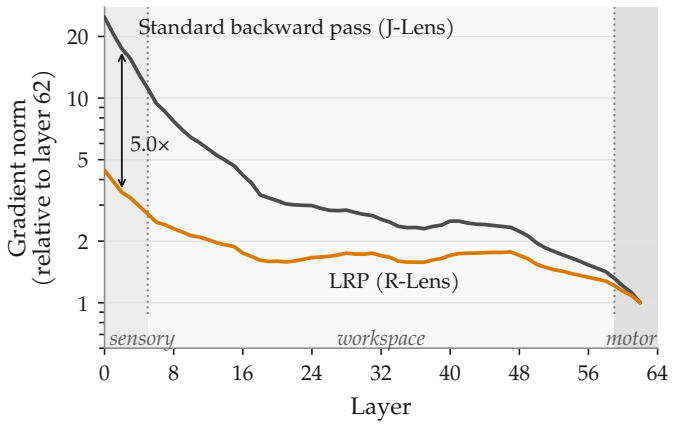
Layer	J-Lens	R-Lens	J++ Lens (ours)
8	␣↵	␣...	␣and
16	␣anyways	␣...	␣Texas
24	␣...	␣...	**␣heart**
32	____	␣...	**␣heart**
40	的心脏	心脏	␣cardiac
48	的心脏	␣cardiac	␣cardiac
56	␣chambers	␣chambers	␣cardiac

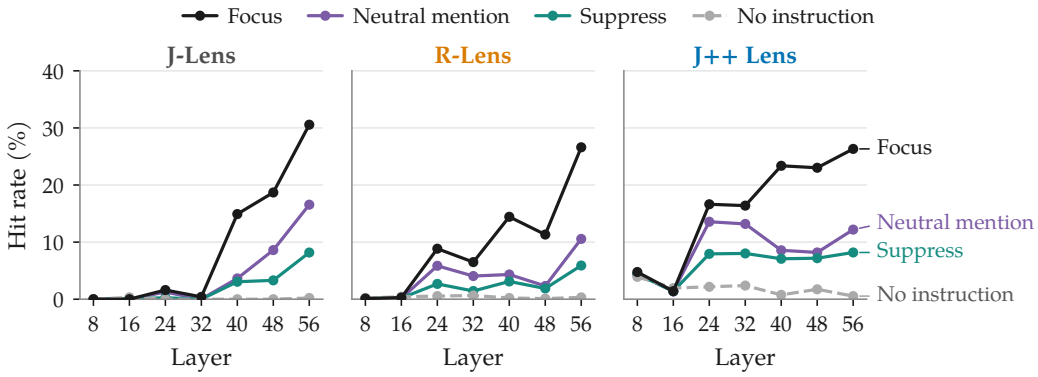
(b)

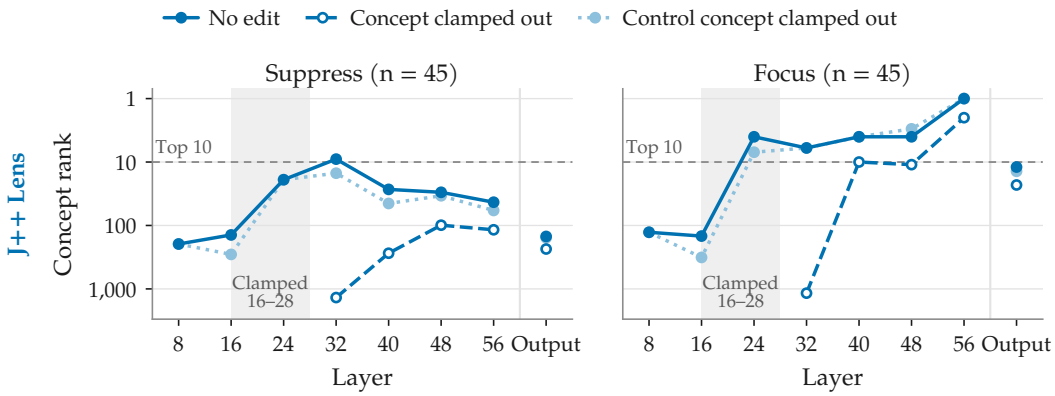

J-Lens
R-Lens
J++ Lens
Recall@10 (%)
100
80
60
40
20
0
Multilingual
Multihop
Typo
Association
Poetry
Recall@10 at layer (%)
50
40
30
20
10
0
0
8
16
24
32
40
48
56
Layer
Swap success (%)
80
60
40
20
0
49.3
50.7
59.7
J
R
J++
Lens

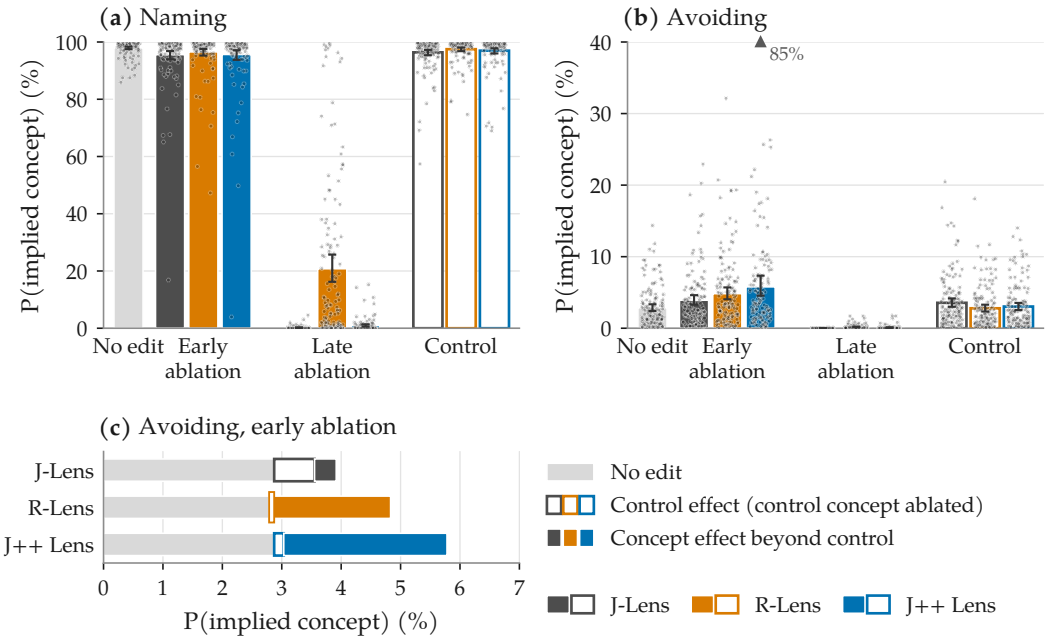
(a) (b) (c)

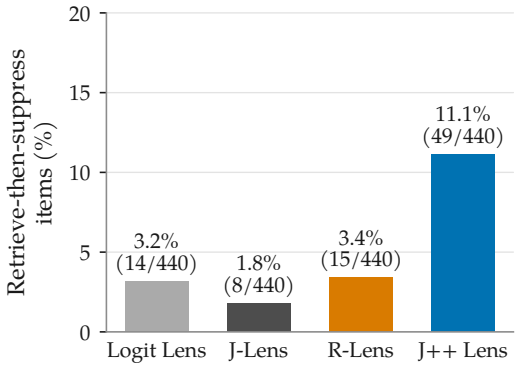

Expert Jacobian applied globally
Best Expert Jacobian
Mean Pooled Jacobian (R-Lens)
J++ Lens
Recall@10 at layer (%)
50
40
30
20
10
0
8
16
24
32
40
48
56
Layer

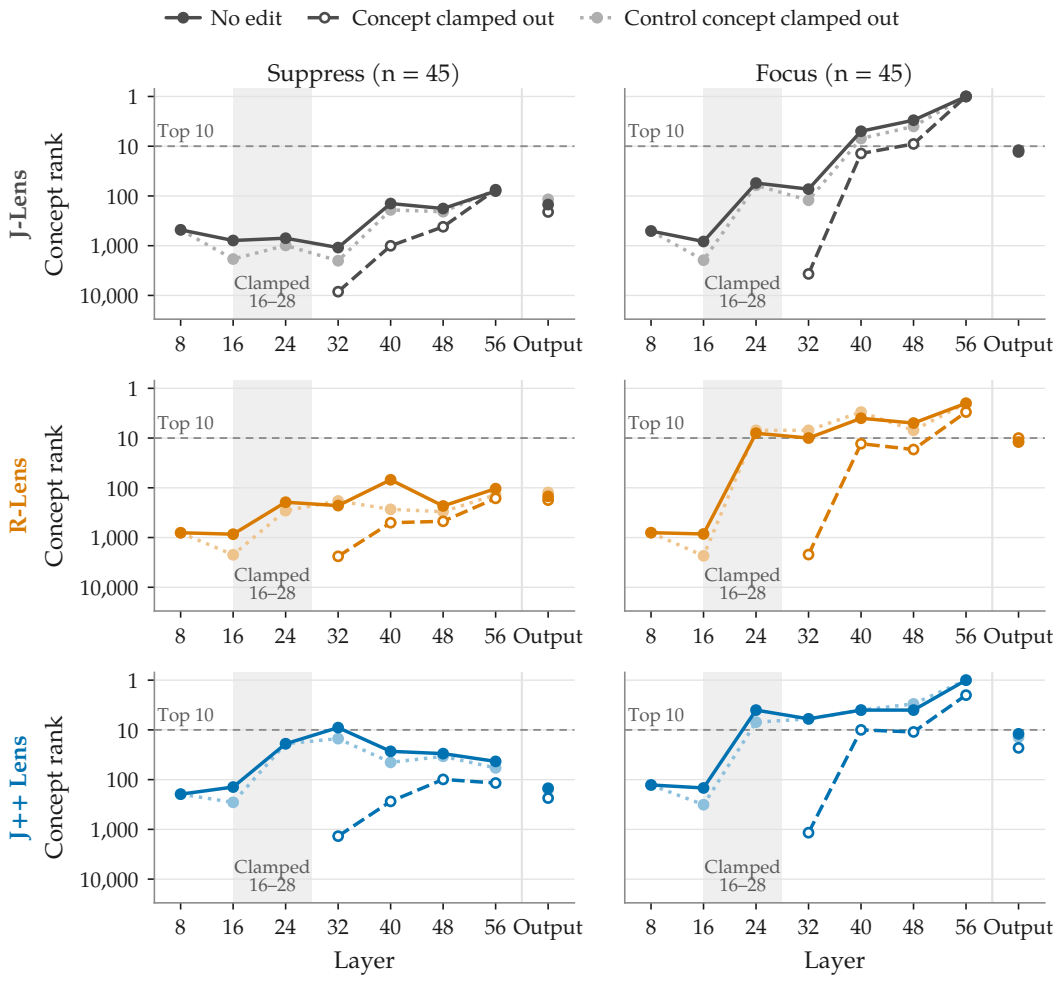

Standard backward pass (J-Lens)
5.0×
LRP (R-Lens)
sensory
workspace
motor
Gradient norm
(relative to layer 62)
20
10
5
2
1
0
8
16
24
32
40
48
56
64
Layer


Focus
Neutral mention
Suppress
No instruction
J-Lens
R-Lens
J++ Lens
Hit rate (%)
0
10
20
30
40
8 16 24 32 40 48 56
Layer
Focus
Neutral mention
Suppress
No instruction


No edit
Concept clamped out
Control concept clamped out
J++ Lens
Concept rank
Suppress (n = 45)
Focus (n = 45)
1
10
100
1,000
Top 10
Clamped
16–28
8
16
24
32
40
48
56
Output
Layer


(a) Naming
P(implied concept) (%)
100
80
60
40
20
0
No edit
Early ablation
Late ablation
Control
(b) Avoiding
P(implied concept) (%)
40
30
20
10
0
85%
No edit
Early ablation
Late ablation
Control
(c) Avoiding, early ablation
J-Lens
R-Lens
J++ Lens
0
1
2
3
4
5
6
7
P(implied concept) (%)
No edit
Control effect (control concept ablated)
Concept effect beyond control
J-Lens
R-Lens
J++ Lens


Retrieve-then-suppress items (%)
20
15
10
5
0
3.2% (14/440)
1.8% (8/440)
3.4% (15/440)
11.1% (49/440)
Logit Lens
J-Lens
R-Lens
J++ Lens


No edit
Concept clamped out
Control concept clamped out
Suppress (n = 45)
Focus (n = 45)
J-Lens
R-Lens
J++ Lens
Concept rank
Top 10
Clamped 16–28
1
10
100
1,000
10,000
8
16
24
32
40
48
56
Output
Layer

Multihop

Fact: The chemical symbol for the element with atomic number 79 is

gold target

Layer	J-Lens			R-Lens			J++ Lens		
8	␣↵	␣...	𝔊	␣...	␣.....	␣,	␣medals	␣bronze	_atomic
16	<br	␣ <	.<	␣...	␣``	␣''	␣and	␣the	␣-
24	Persons	↵↵	atoms	atomic	原子	atom	␣atomic	␣Atomic	atom
32	␣_____	_____	␣_____	␣_____	␣<	name	2	1	0
40	␣metals	符号	abbrev	符号	␣metals	␣atomic	␣metals	␣metal	␣atomic
48	␣	_____	___	_____	___	__	1	2	0
56	?\	⇥	是什么	?\	?	.\	7	metal	symbol

Multilingual

O oposto de "cima" é "

Portuguese **opposite** **up** **down** target

Layer	J-Lens			R-Lens			J++ Lens		
8	ness	␣□□	enticati…	␣"	␣..."	..."	␣French	␣Cinder	**␣Portugu…**
16	ness	␣□□	s	␣"	"	s	␣Arizona	␣Kaz	␣Al
24	ness	s	␣	␣"	"	␣,	**␣Portugu…**	␣Brazili…	**␣opposite**
32	␣	_____	␣_____	␣"	...	␣	**␣Portugu…**	␣or	␣Spanish
40	_____	␣	␣lowerca…	...	"	\"	␣bottom	␣below	**␣down**
48	_____	___	␣_____	_____	"	␣__	␣bottom	**␣opposite**	␣or
56	␣bottom	␣below	bottom	␣bottom	bottom	␣below	␣bottom	bottom	␣below

Typo

They moved into the new hosue

house target other form of the target

Layer	J-Lens			R-Lens			J++ Lens		
8	ö	erder	eat	␣,	␣of	␣and	**␣house**	␣houses	**house**
16	␣alot	ö	␣occured	␣alot	␣teh	␣occured	**␣House**	␣HOUSE	**house**
24	ö	␣,	i	e	i	␣,	**house**	**␣House**	**␣house**
32	_____	___	␣___	(	i	e	**␣House**	**house**	**␣house**
40	_____	**␣house**	(	(	i	a	**␣house**	**house**	h
48	_____	___	_____	_____	(	___	␣new	**␣house**	(new
56	␣yesterd…	句子	昨天	(	␣yesterd…	,	␣yesterd…	␣after	␣without

Association

The bench had been there longer than she had, and the brass plaque said it had been someone's favorite spot before it was hers.

park target

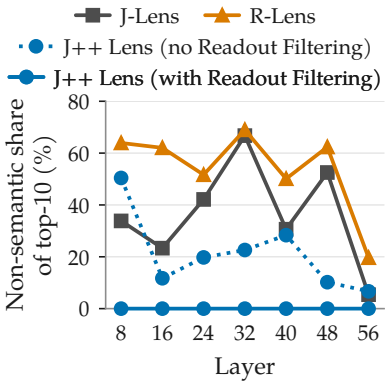
Layer	J-Lens			R-Lens			J++ Lens		
8	␣□□	␣folks	␣jewelry	␣just	␣only	␣that	␣jewelry	␣neighbo…	␣someday
16	␣folks	␣pretty	␣really	␣pretty	␣somethi…	␣↵	␣Texas	␣Houston	␣Oklahoma
24	␣↵↵	␣↵↵	–but	␣somethi…	␣just	␣somehow	␣stories	␣old	␣somethi…
32	_____	␣_____	␣somehow	␣↵	␣	␣somethi…	␣old	␣stories	␣-
40	-	␣	–but	␣↵	␣	␣someone	␣old	␣name	␣history
48	␣someone	_____	␣somehow	_____	␣someone	<|im_end…	␣old	␣someone	␣stories
56	␣rewrite	␣she	改写	<|im_end…	␣she	␣Before	␣Before	␣before	␣she

Poetry

A rhyming couplet: / She gathered shells along the empty shore, / And wished the tide would bring her something

more target

Layer	J-Lens			R-Lens			J++ Lens		
8	ly	“	␣	␣	--	a	␣lyric	␣poetic	这首
16	``	什么意思	什么	␣_____	``	␣``	␣poetic	␣poetry	␣Song
24	␣↵↵	␣↵	ly	↵	␣poetic	␣poetry	␣poet	␣poetic	␣poetry
32	_____	___	␣_____	_____	↵	␣↵	␣poem	␣poetry	␣poetic
40	_____	␣poetic	␣poetry	␣↵	↵	_____	␣poem	␣poetic	␣poetry
48	_____	___	␣_____	_____	↵	↵↵	␣The	␣poet	␣poem
56	␣until	Then	␣then	␣Then	Then	␣then	␣until	␣while	␣Until



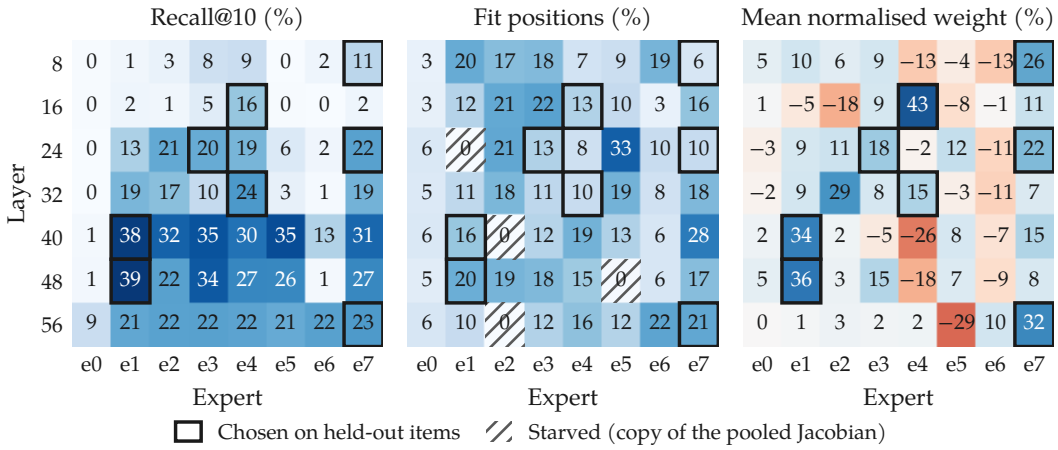
(a)

They moved into the new hosue

house target other form of the target

Layer	J-Lens			J++ Lens (no Readout Filtering)		
8	ö	erder	eat	**␣house**	␣houses	**house**
16	␣alot	ö	␣occured	**␣House**	**house**	␣HOUSE
24	ö	␣,	i	**house**	**␣House**	**␣house**
32	____	___	␣___	**␣House**	**house**	**␣house**
40	____	**␣house**	(	**␣house**	**house**	h
48	____	___	_____	␣new	**␣house**	new
56	␣yester…	句子	昨天	␣yester…	.	␣after

(b)


Recall@10 (%)
Fit positions (%)
Mean normalised weight (%)
Layer
8 0 1 3 8 9 0 2 11
16 0 2 1 5 16 0 0 2
24 0 13 21 20 19 6 2 22
32 0 19 17 10 24 3 1 19
40 1 38 32 35 30 35 13 31
48 1 39 22 34 27 26 1 27
56 9 21 22 22 22 21 22 23
3 20 17 18 7 9 19 6
3 12 21 22 13 10 3 16
6 0 21 13 8 33 10 10
5 11 18 11 10 19 8 18
6 16 0 12 19 13 6 28
5 20 19 18 15 0 6 17
6 10 0 12 16 12 22 21
5 10 6 9 −13 −4 −13 26
1 −5 −18 9 43 −8 −1 11
−3 9 11 18 −2 12 −11 22
−2 9 29 8 15 −3 −11 7
2 34 2 −5 −26 8 −7 15
5 36 3 15 −18 7 −9 8
0 1 3 2 2 −29 10 32
e0 e1 e2 e3 e4 e5 e6 e7
Expert
Chosen on held-out items
Starved (copy of the pooled Jacobian)

Best expert at layer 16
Formatting expert at layer 16

... the Meridian Hall . These were created by Ignazio Danti , the papal cosmographer , in association with the Gregorian Calendar Reform . The ...

... Sami Lepisto and Mike Commodore . Hejda , who played four of his first five NHL seasons with the Blue Jackets , was ...

... Wales , and The Parable of the Great Supper (1934) for St. George 's Chapel , Waddon . ...

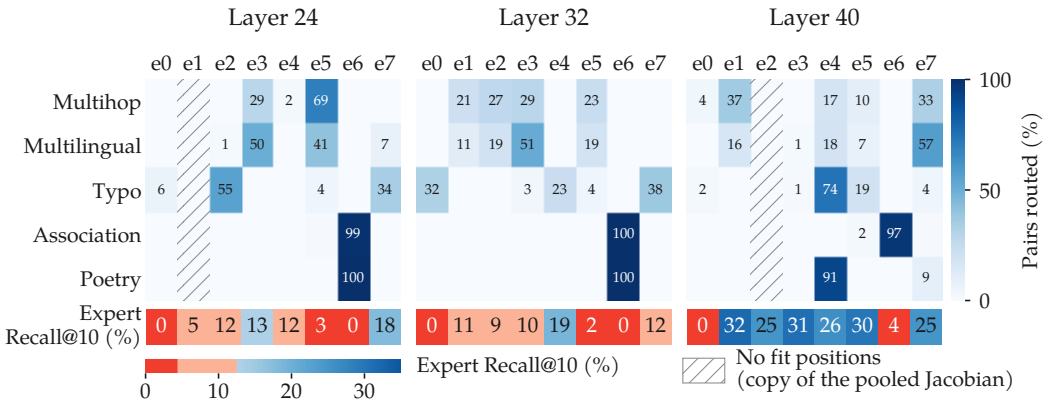
... Second United States Artillery , consisting of sixty @-@ five men , was transferred to Little Rock under the command of Captain James Totten . On ...

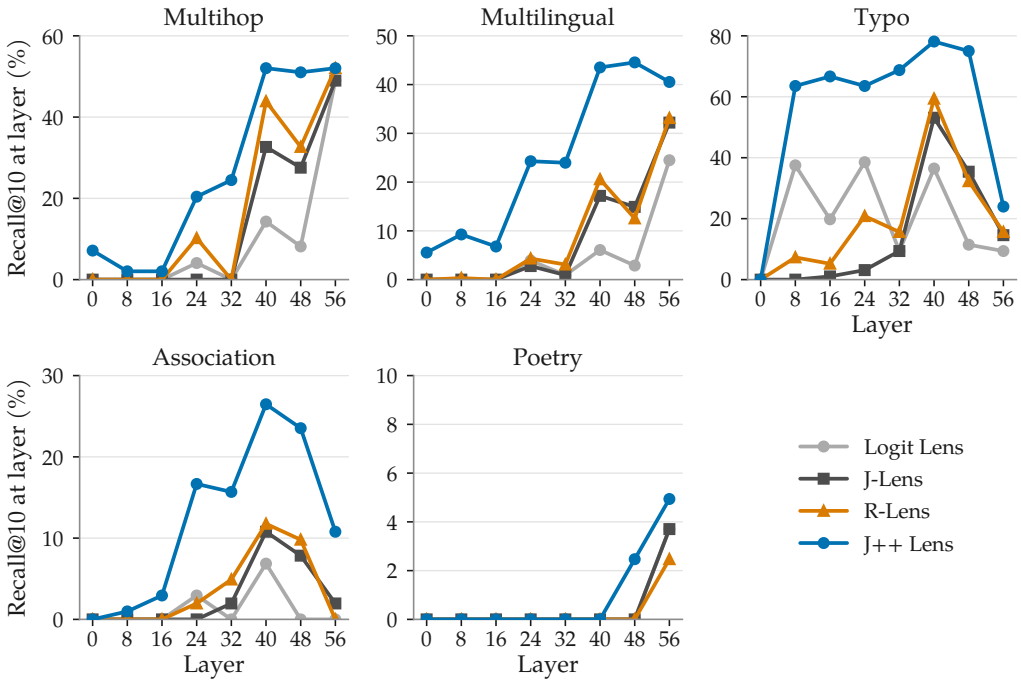
(a)

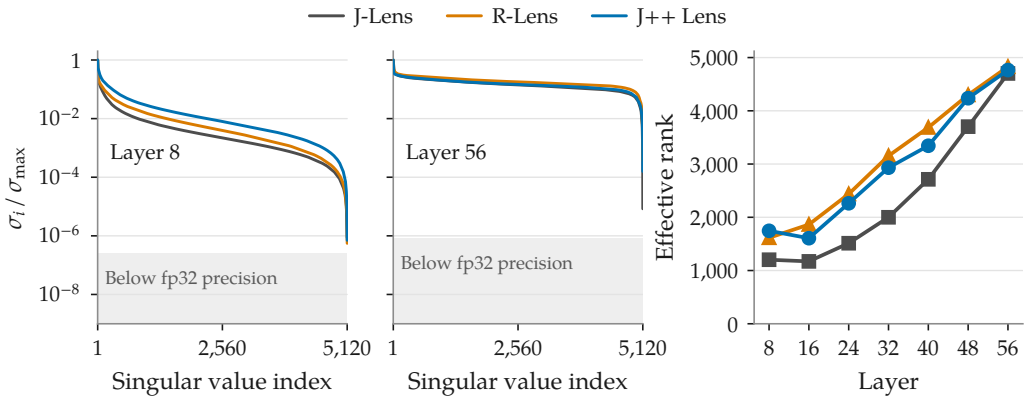
Expert	Recall@10 (%)	Weight (%)	Content word	Sub-word	Function word	Digit	Punc-tuation	Mark-up	Non-semantic	Most over-represented tokens
Chosen on held-out items († by one of the two fold lenses)										
L8 e7	11.3	26	0.8	8.1	<0.1	0.8	<0.1	0	<0.1	`y 2 s`
L16 e4	15.7	43	1.5	4.1	0.1	0.5	0.1	0	0.1	`␣Valk y 3`
L24 e3†	20.4	18	0.2	0.5	1.9	3.3	0.3	0	0.5	`␣the 9 1 8 2`
L24 e7†	22.3	22	1.4	3.9	0.2	1.0	0.2	0.1	0.2	`y 3 1 2`
L32 e4	23.7	15	1.6	4.1	<0.1	0.8	0	0	0	`y ␣Valk 3`
L40 e1	37.9	34	0.1	0.4	2.2	2.6	0.3	<0.1	0.6	`␣the ␣by 1 ␣of 9`
L48 e1	38.6	36	0.1	0.3	2.9	0.1	0.3	0	0.3	`␣at ␣by ␣a ␣on ␣from`
L56 e7	23.5	32	1.3	1.8	0.7	1.1	0.4	0.6	0.5	`y ␣Valk ␣an 3 5`
Formatting experts (at least half non-semantic positions)										
L8 e0	0.2	5	0	0	0	0	2.6	33	6.5	`␣@ - @`
L8 e5	0.0	−4	0	0	0	0	9.2	0	6.5	`␣, ␣. ␣' ␣) ␣"`
L16 e0	0.2	1	0	0	0	0.3	2.2	31	6.4	`␣@ @ - ␣ 0`
L16 e5	0.3	−8	0.1	<0.1	1.1	0.3	5.6	0	3.9	`␣The ␣, ␣. ␣( ␣"`
L16 e6	0.0	−1	<0.1	0	0	0	8.1	0	6.5	`␣' ␣) ␣. ␣, ␣`
L24 e0	0.0	−3	<0.1	0	0.1	0.4	4.6	16	6.1	`␣@ @ - ␣' ␣)`
L24 e6	1.7	−11	0.1	0.1	1.0	0.2	5.7	0	4.0	`␣The ␣. ␣, ␣) ␣"`
L32 e0	0.0	−2	<0.1	0	0.1	0.7	4.5	15	6.0	`␣@ - ␣' @ ␣)`
L32 e6	0.5	−11	0.1	<0.1	0.9	0.2	6.2	0.2	4.3	`␣. ␣) ␣The ␣, ␣"`
L40 e0	0.8	2	<0.1	<0.1	0.1	0.8	4.1	17	6.0	`␣@ @ - ␣' ␣)`
L40 e6	12.6	−7	<0.1	0	0.1	0	9.1	0	6.3	`␣. ␣, ␣) ␣" ␣'`
L48 e0	1.0	5	<0.1	<0.1	0.1	1.1	2.6	22	5.9	`␣@ @ - ␣' ␣`
L48 e6	0.7	−9	<0.1	0	<0.1	<0.1	9.3	0	6.5	`␣. ␣, ␣) ␣"`
L56 e0	9.1	0	<0.1	0	<0.1	0.5	6.2	12	6.3	`␣@ ␣. - ␣' @`
Worst remaining experts (lowest recall@10 at the layer, excluding formatting experts)										
L8 e1	1.3	10	0.1	0.2	1.8	3.6	0.4	0	1.0	`␣the ␣a ␣The ␣( ␣an`
L16 e2	1.4	−18	0.1	0.2	2.8	<0.1	0.3	0	0.4	`␣at ␣had ␣for ␣that ␣from`
L24 e5	5.6	12	0.8	0.3	1.8	0.6	0.3	0.1	0.4	`␣had ␣for ␣at ␣from ␣that`
L32 e5	3.3	−3	1.2	0.4	1.3	0.2	0.4	0	0.4	`␣was ␣had ␣were ␣that ␣is`
L40 e4	29.7	−26	1.7	1.4	0.2	2.3	0.2	<0.1	0.3	`␣game 3 4 ␣( 0`
L48 e2	22.3	3	1.0	0.3	1.7	<0.1	0.4	0	0.3	`␣was ␣had ␣is ␣were ␣The`
L56 e5	20.6	−29	1.8	1.7	0.1	1.4	0.5	0	0.3	`␣) ␣game 3 8 0`

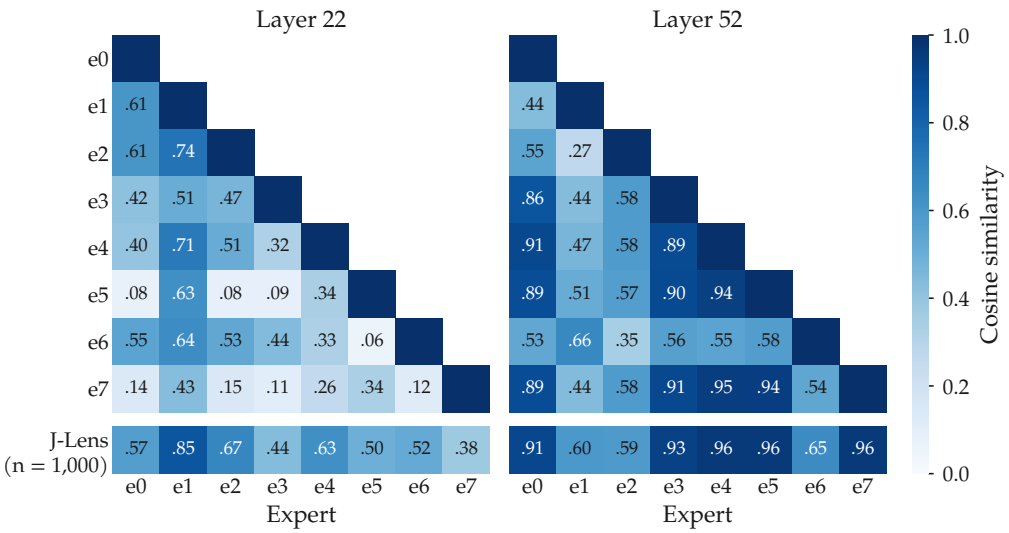
Lift (×), log scale ≤1/3 1 2 5 ≥10 — 0 no position in the class

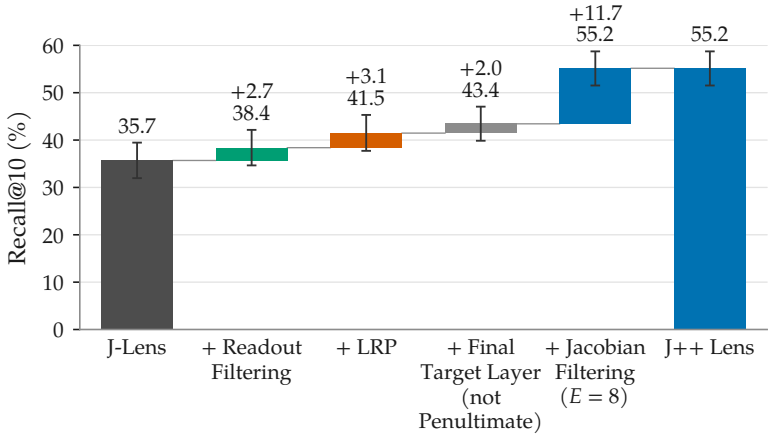
(b)

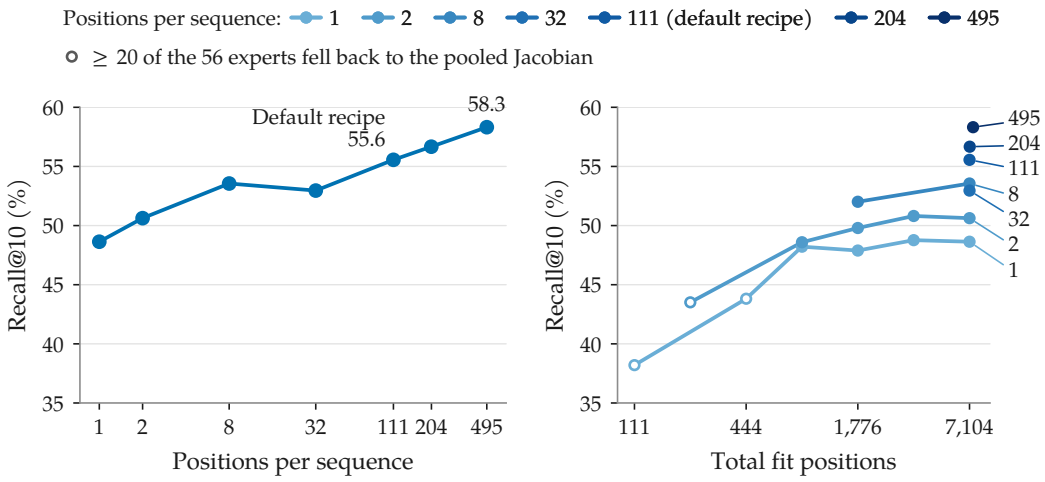

Layer 24
Layer 32
Layer 40
e0 e1 e2 e3 e4 e5 e6 e7
Multihop
Multilingual
Typo
Association
Poetry
Expert Recall@10 (%)
0 5 12 13 12 3 0 18
0 11 9 10 19 2 0 12
0 32 25 31 26 30 4 25
0 10 20 30
Expert Recall@10 (%)
Pairs routed (%)
100
50
0
No fit positions
(copy of the pooled Jacobian)


Multihop
Multilingual
Typo
Association
Poetry
Recall@10 at layer (%)
Layer
Logit Lens
J-Lens
R-Lens
J++ Lens


J-Lens
R-Lens
J++ Lens
$\sigma_i / \sigma_{\max}$
1
10^{-2}
10^{-4}
10^{-6}
10^{-8}
Layer 8
Below fp32 precision
1
2,560
5,120
Singular value index
Layer 56
Below fp32 precision
1
2,560
5,120
Singular value index
Effective rank
5,000
4,000
3,000
2,000
1,000
0
8
16
24
32
40
48
56
Layer


Layer 22
Layer 52
e0
e1 .61
e2 .61 .74
e3 .42 .51 .47
e4 .40 .71 .51 .32
e5 .08 .63 .08 .09 .34
e6 .55 .64 .53 .44 .33 .06
e7 .14 .43 .15 .11 .26 .34 .12
J-Lens (n = 1,000) .57 .85 .67 .44 .63 .50 .52 .38
e0 e1 e2 e3 e4 e5 e6 e7
Expert
e0
e1 .44
e2 .55 .27
e3 .86 .44 .58
e4 .91 .47 .58 .89
e5 .89 .51 .57 .90 .94
e6 .53 .66 .35 .56 .55 .58
e7 .89 .44 .58 .91 .95 .94 .54
J-Lens (n = 1,000) .91 .60 .59 .93 .96 .96 .65 .96
e0 e1 e2 e3 e4 e5 e6 e7
Expert
Cosine similarity
1.0
0.8
0.6
0.4
0.2
0.0


Recall@10 (%)
60
50
40
30
20
10
0
35.7
+2.7
38.4
+3.1
41.5
+2.0
43.4
+11.7
55.2
55.2
J-Lens
+ Readout Filtering
+ LRP
+ Final Target Layer (not Penultimate)
+ Jacobian Filtering ($E = 8$)
J++ Lens


Positions per sequence: 1 2 8 32 111 (default recipe) 204 495
≥ 20 of the 56 experts fell back to the pooled Jacobian
Recall@10 (%)
Default recipe
55.6
58.3
Positions per sequence
(a)
Total fit positions
(b)
495
204
111
8
32
2
1